Problem 16.1.1

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{\cos z}{z^4}$$

Solution

The Taylor series for $\cos z$ is,

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n},$$

Dividing by z^4,

$$\boxed{\frac{\cos z}{z^4} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n-4}}$$

The Taylor series for $\cos z$ converges with $R = \infty$, so the Laurent series converges for every $z \neq 0$, or $0 < |z| < \infty$.

Problem 16.1.3

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{\exp z^2}{z^3}$$

Solution

The Taylor series for $\exp z^2$ is,

$$\exp z^2 = \sum_{n=0}^{\infty} \frac{z^{2n}}{n!}$$

Dividing by z^3,

$$\boxed{\frac{\exp z^2}{z^3} = \sum_{n=0}^{\infty} \frac{z^{2n-3}}{n!}}$$

The Taylor series for $\exp z^2$ converges with $R = \infty$, so the Laurent series converges for every $z \neq 0$, or $0 < |z| < \infty$.

Problem 16.1.5

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{1}{z^2 - z^3}$$

Solution

$$\frac{1}{z^2 - z^3} = \frac{1}{z^2}\frac{1}{1-z}$$

The Taylor series for the geometric series $\frac{1}{1-z}$ is,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

Dividing by z^2,

$$\boxed{\frac{1}{z^2 - z^3} = \sum_{n=0}^{\infty} z^{n-2}}$$

The Taylor series for $\frac{1}{1-z}$ converges on $|z| < 1$, so the Laurent series converges on $0 < |z| < 1$.

Problem 16.1.8

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{e^z}{z^2 - z^3}$$

Solution

The Taylor series for e^z is,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Multiplying by the series from **16.1.5**,

$$\frac{e^z}{z^2 - z^3} = \sum_{n=0}^{\infty} \frac{z^{2n-2}}{n!}$$

The Taylor series for e^z converges everywhere. As in **16.1.5**, the Laurent series converges on $0 < |z| < 1$.

Problem 16.1.13

Find the Laurent series that converges for $0 < |z - i| < R$ and determine the precise region of convergence.

$$\frac{1}{z^3 \left(z - i\right)^2}$$

Solution

Problem 16.1.22

Find all Taylor and Laurent series with center $z_0 = i$. Determine the precise regions of convergence.

$$\frac{1}{z^2}$$

Solution

Problem 16.1.23

Find all Taylor and Laurent series with center $z_0 = 0$. Determine the precise regions of convergence.

$$\frac{z^8}{1 - z^4}$$

Solution

Problem 16.2.1

Determine the location and order of the zeros.

$$\sin^4 \frac{1}{2} z$$

Solution

Problem 16.2.3

Determine the location and order of the zeros.

$$(z + 81i)^4$$

Solution

Problem 16.2.5

Determine the location and order of the zeros.

$$z^{-2} \sin^2 \pi z$$

Solution

Problem 16.2.7

Determine the location and order of the zeros.

$$z^4 + (1 - 8i)z^2 - 8i$$

Solution

Problem 16.2.9

Determine the location and order of the zeros.

$$\sin 2z \cos 2z$$

Solution

Problem 16.2.13

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{1}{(z+2i)^2} - \frac{z}{z-i} + \frac{z+1}{(z-i)^2}$$

Solution

Problem 16.2.15

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$z \exp\left(\frac{1}{(z-1-i)^2}\right)$$

Solution

Problem 16.2.17

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\cot^4 z$$

Solution

Problem 16.2.19

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{1}{e^z - e^{2z}}$$

Solution

Problem 16.2.21

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{e^{\frac{1}{z-1}}}{e^z - 1}$$

Solution

Problem 16.3.3

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{\sin 2z}{z^6}$$

Solution

Problem 16.3.5

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{8}{1+z^2}$$

Solution

Problem 16.3.7

Find all the singularities in the finite plane and the corresponding residues.

$$\cot \pi z$$

Solution

Problem 16.3.9

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{1}{1 - e^z}$$

Solution

Problem 16.3.11

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{e^z}{(z-\pi i)^3}$$

Solution